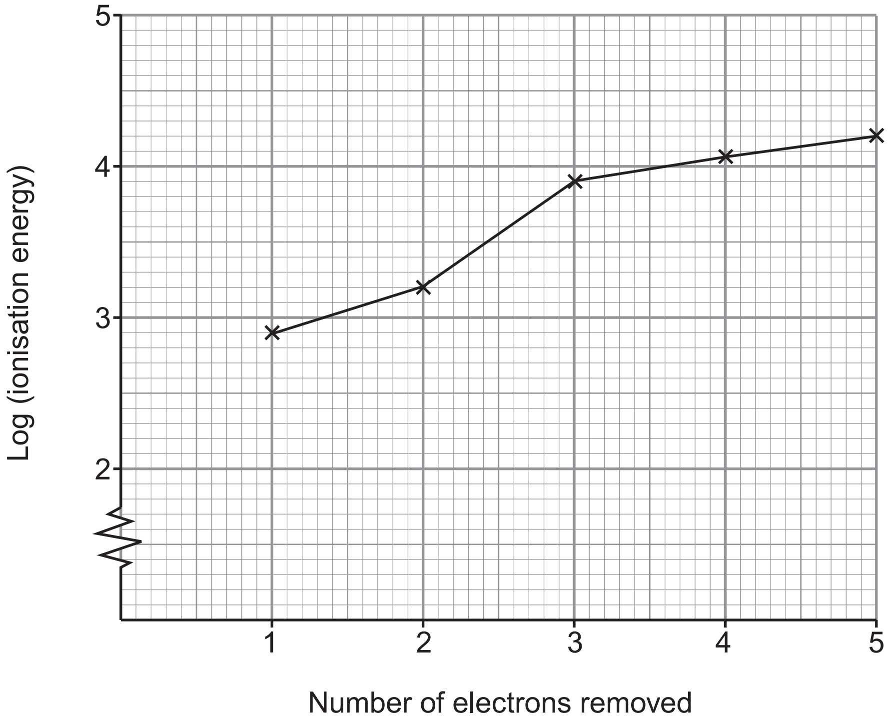
11. Two students were discussing how electrons affect ionisation energies and the shape of molecules.

(a) Elements **X**, **Y** and **Z** are in the same period of the Periodic Table. The graph below shows the first five ionisation energies of element **X**.



(i) State how the graph shows that element **X** is in Group 2. [1]

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(ii) Element **Y** has an atomic number one less than **X**, while element **Z** has an atomic number one more than **X**.

Since there is a general increase in first ionisation energies across a period, one student said:

"Element **X**'s first ionisation energy will be higher than that of element **Y** and lower than that of element **Z**."

Is he correct? Justify your answer by referring to both parts of his statement. [3]

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(b) (i) Write an equation to represent the second ionisation energy of sodium. Include state symbols. [1]

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(ii) The second ionisation energy of sodium is 4560 $kJ\,mol^{-1}$.

Calculate the value of the wavelength, in nm, of the radiation that corresponds to this energy change. [4]

Wavelength = ... nm

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(c) The second student said:

"When you consider CH_4, $AlCl_3$ and $BeCl_2$, the bond angle between the atoms in the molecule decreases as the number of bonds increases.

Therefore, the greater the number of bonds in any molecule the smaller the bond angle."

(i) State the bond angles in the molecules given above. [3]

$H - \hat{C} - H$ .... °

$Cl - \hat{Al} - Cl$ .... °

$Cl - \hat{Be} - Cl$ .... °

(ii) Use your knowledge of VSEPR theory to explain why the second sentence is incorrect. Give an example to support your answer. [2]

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